Polynomials

1 Determine whether each of the following is a polynomial:

a $2x^3 - 4x^5 + 3$

b $\sqrt{2}x^6 + 7x$

c $3x^3 + \dfrac{2}{x^7} - 1$

d $\dfrac{3x^3 - 4x^2 + 2}{\pi^2 + 1}$

e $4x^{\frac{1}{4}} + 2x^5 + 2$

f $\left(x^3 + 3\right)\left(\sqrt{7} + 2x^{-2}\right)$

g $3x^7 + x^3 - 7\sqrt{x} - 12$

h $-7a^9 - 10ab + 6b^2$

2 For the following polynomials, state:

i The degree.

ii The leading coefficient.

iii The constant term.

a $P\left(x\right) = 2x^7 + 2x^5 + 2x + 2$

b $P\left(x\right) = 7\sqrt{6} - \sqrt{5}x^5 + 5$

c $P\left(x\right) = \dfrac{x^7}{5} + \dfrac{x^6}{6} + 5$

d $P(x) = 6 - \dfrac{7}{10}x^3$

Add and subtract polynomials

3 $R\left(x\right)$ is the result of adding polynomials $P\left(x\right)$ and $Q\left(x\right)$, where $P\left(x\right) = 3x - 6x^2 + 7$ and $Q\left(x\right) = 2x^3 - 4x - 3$.

a Express $R\left(x\right)$ in simplest form.

b Determine the degree of $R\left(x\right)$.

c Identify the leading coefficient of $R\left(x\right)$.

d State whether $R\left(x\right)$ is a polynomial.

4 Consider $P\left(x\right) = 5x^2 - 6x - 4$ and $Q\left(x\right) = -2x + 6$.
Form a simplified expression for $P\left(x\right) + Q\left(x\right)$.

5 Consider $P\left(x\right) = 3x^2 + 7x - 6$ and $Q\left(x\right) = 6x - 7$.
Form a simplified expression for $P\left(x\right) - Q\left(x\right)$.

6 Consider $A\left(x\right) = -6x^2 - 6$, $B\left(x\right) = 4x - 7$ and $C\left(x\right) = x^2 - 5x$.
Form a simplified expression for $B\left(x\right) - A\left(x\right)$.

7 Consider $A\left(x\right) = -2x^2 - 3$, $B\left(x\right) = -6x + 3$ and $C\left(x\right) = 6x^2 + 2x$.
Form a simplified expression for $A\left(x\right) + B\left(x\right) + C\left(x\right)$.

8 Consider $A(x) = 5x^2 + 2$, $B(x) = -3x + 3$ [redacted] $(x) = -2x^2 + 7x$.
Form a simplified expression for $A(x) - B(x) - C(x)$.

9 Consider $P(x) = 3x^3 - 7x^2 - x + 2$, $Q(x) = 3 - 9x^3$ and $R(x) = -3x^2 + 9x - x^3 + 6$.

a Find $P(x) - R(x)$ in simplest form.

b Find $P(x) - (R(x) - Q(x))$ in simplest form.

10 Simplify:

a $\left(-8x^2 - 3\right) + \left(-4x^2 - 8x\right)$

b $\left(3x^3 - 9x^2 - 8x - 7\right) + \left(-7x^3 - 9x\right)$

c $\left(\frac{1}{2}x^2 - \frac{1}{4}x + \frac{1}{2}\right) + \left(\frac{1}{3}x^2 + \frac{2}{5}x - \frac{1}{4}\right)$

d $\left(3x^3 - 3x^2 - 8\right) + \left(-5x^3 + 9x^2 + 1\right)$

e $3(-2x + 7) - \left(7x^2 + 8x - 5\right) + 7x$

f $\left(-6x^3 + 4x^2 + 4x\right) - \left(3x^3 + x + 1\right)$

g $\left(-5x^2 + 6x - 4\right) - \left(-7x^2 + 7x + 9\right)$

h $\left(-2x^3 + 7x^2 + 9x - 2\right) - \left(9x^3 + 7x^2\right)$

i $\left(-x^3 - 3x^2 - 4\right) - \left(x^3 - 9x - 9\right)$

j $\left(x^3 - 4x^2 + 4\right) + \left(8x^2 - 2\right) - \left(-2x^3 + 4x^2 + 8x + 9\right)$

11 Let $P(x) = ax^2 + (b - 5)x - 1$ and $Q(x) = x^2 - 5x + 2$.
If $R(x) = 2x^2 + 2x + 1$ is the result of adding $P(x)$ and $Q(x)$, find the values of a and b.

Multiply polynomials

12 Simplify:

a $(x+2)(x+5)$

b $(x-2)(x-3)$

c $(a+9)(b-6)$

d $(m-6)(m+3)$

e $(5m+8)(m-3)$

f $(7y-6)(6y+6)$

g $(7w+5)(5w+2)$

h $\left(\frac{1}{3}y+\frac{1}{4}\right)\left(\frac{1}{3}y-\frac{1}{6}\right)$

i $(9y-8)(-10y-6)$

j $9(y+8)(y+2)$

k $-(x+5)(x+2)$

l $-5(y+4)(y+5)$

m $-4(y-7)(y+4)$

n $5(3x-2)(-5x+4)$

13 Simplify:

a $6(-5x+3)(x+2)-2$

b $(2n+5)(5n+2)-4$

c $(a+2)\left(5a^2-2a+2\right)$

d $(u-4)\left(5u+5+u^2\right)$

e $(5a+5)\left(-5a-4-5a^2\right)$

f $\left(5c^2-2\right)\left(4c^2+2-3c\right)$

g $\left(-3m^2-2m+2\right)(4m+4)$

h $(-7x+3y+9)(8x+12)$

14 Consider the statement $(u+1)\left(u^2-u+1\right)=u^3+1$.
Is this statement true or false?

15 Explain how you can find the coefficient of x in the following expression without fully distributing the multiplication.

$$\left(x^2-3x-5\right)(5x-4)$$

16 Consider $R(x)$, the product of $P(x)=3x^5-3$ and $Q(x)=-2x^7+5x^5+6$.

a Find the degree of $R(x)$.

b Find the constant term of $R(x)$.

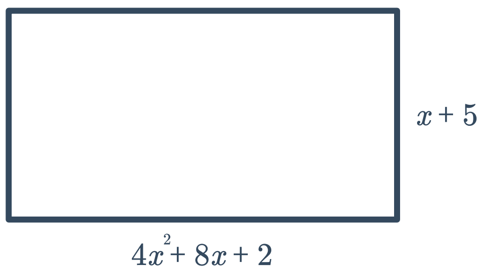
c Is $R(x)$ a polynomial?

17 Express $(x+3)^2+2(x+3)-3$ as the product of two binomials.

Applications

18 Find a simplified polynomial that represents the perimeter of the following figures:

a



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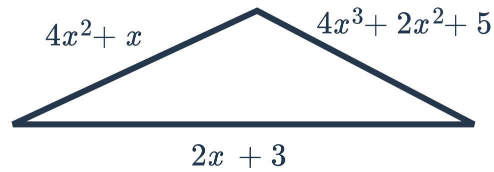

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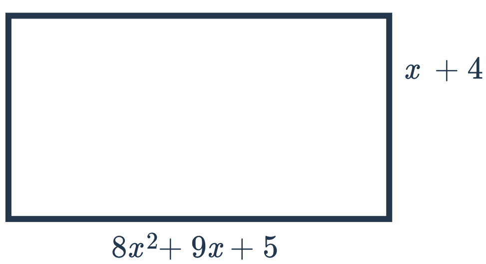

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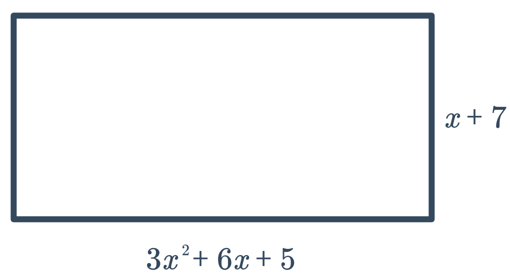

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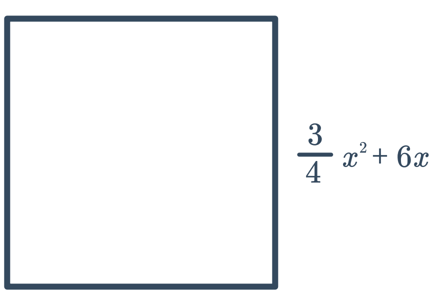

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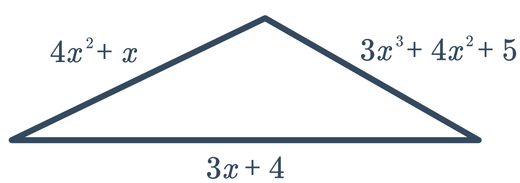
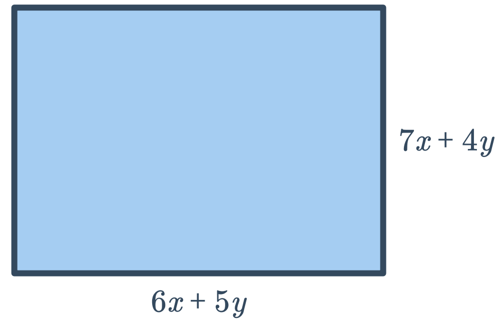


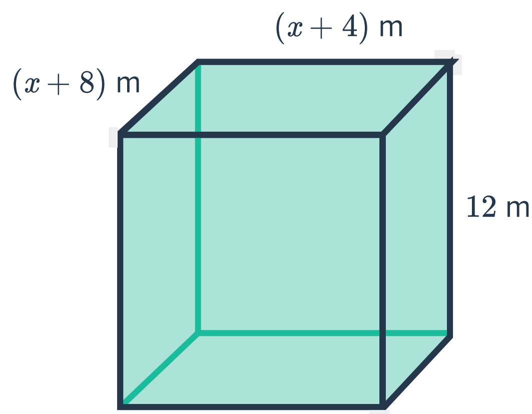
19 A picture frame has a length of $8x^2 - 9x + 3$ and a width of $6x^3 + 9x^2$. Form a fully simplified expression for the perimeter of the rectangular picture frame.

20 Form a fully simplified polynomial expression for the area of the given rectangle.

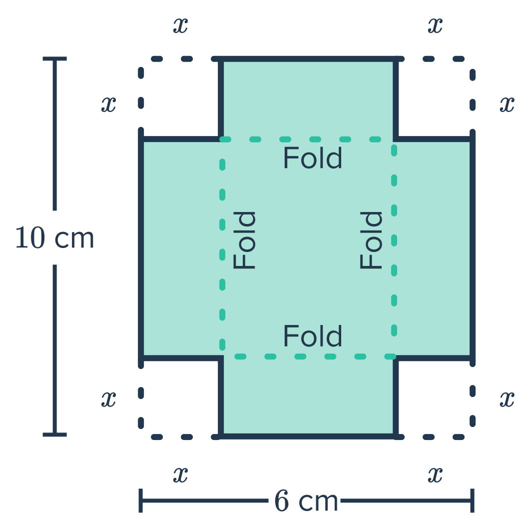


21 A landscape design company makes large raised boxes for plantings. The height of the boxes they build is $2y + 3$ ft and the area of the rectangular base measures $4y^2 + 5y + 1$ ft^2. If the box is filled with soil, what is the volume of soil needed? Give your answer as a fully simplified polynomial.

22 Consider the rectangular prism shown with its dimensions labeled:
Form a simplified polynomial expression for the volume of the given prism.



23 Consider a rectangular cardboard measuring 10 cm by 6 cm. It is to be converted into a box with no lid by cutting out square corners measuring x cm in length and folding up the sides.
Form an expression for the volume of the cardboard box in terms of x. Give your answer as a simplified polynomial.

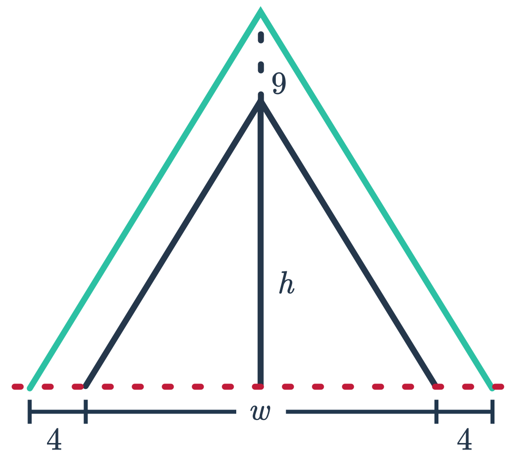


24 Elizabeth wants to know the difference in cross-sectional area of the door of her tent and the tent itself. The proportions of the tent are exactly the same as the door, as seen in the picture. The triangular door has a length and height of w and h respectively. The top of the tent is 9 in greater than the door, and its ends are 4 in longer than the ends of the door.

a Express the height of the tent in terms of h.

b Express the base length of the tent in terms of w.

c Now, find a simplified expression for the difference between the cross-sectional area of the tent and the tent door.



25 Han is trying to determine the dimensions of a rectangular box that will result in the largest volume. The height of the box is 4 ft, the width is $4y\text{ ft}$ and the length is 3 ft more than double the width.
Write a simplified expression for the volume of the box.